SIG788 – Engineering AI Solutions

Task 5 D: Object Detection and Tracking using Azure Computer Vision

Student Name: Vijay Ashley Rodrigues Kinnigoly

Student ID: 225704862

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1. Introduction

A branch of artificial intelligence called computer vision makes it possible for machines to decipher and evaluate visual data from pictures and movies. Object detection is one of the most used computer vision techniques. It locates and recognizes objects in an image by labelling them and creating bounding boxes around them. Applications including driverless cars, retail analytics, smart environments, and surveillance systems frequently use object detection.

Developers may access robust computer vision capabilities without creating machine learning models from scratch thanks to cloud-based AI platforms. Azure AI Vision is one such platform that offers capabilities for object detection and image analysis.

Using Azure AI Vision via the Vision Studio interface, object detection studies were carried out in this work. The effectiveness of the item detection algorithm was assessed by analyzing several photos with commonplace objects. The purpose of the studies was to determine the system's object detection accuracy as well as any potential model constraints while analyzing real-world photos.

2. Methodology

Azure AI Vision was used for the object detection investigations using the Vision Studio interface. Access to the platform's numerous machine vision capabilities was made possible via the creation of an Azure AI Services resource in the Azure portal.

Pictures were analyzed and objects were found using Vision Studio's "Detect common objects in pictures" tool. After each image was submitted to the interface, the system processed it automatically and created bounding boxes around any objects it found. Each detection was accompanied by a confidence score that indicated the model's level of certainty regarding the classification.

For the experiment, three distinct photos were chosen. To assess the system's performance under various circumstances, these pictures depict several object detection scenarios.

Desk and workspace settings were chosen because they frequently include a variety of commonplace items such tables, seats, keyboards, monitors, and plants. Due to the near proximity of several things in the scene, these surroundings offer an excellent test case for object detection algorithms.

The author's personal workstation layout is depicted in one of the experiment's photographs, while the other two are from publicly accessible stock photo sources. The system was evaluated in a variety of real-world situations by using both external and personal photos.

Additionally, the system produced JSON output that included confidence scores and bounding box coordinates for every object it found.

3. Object Detection Experiments

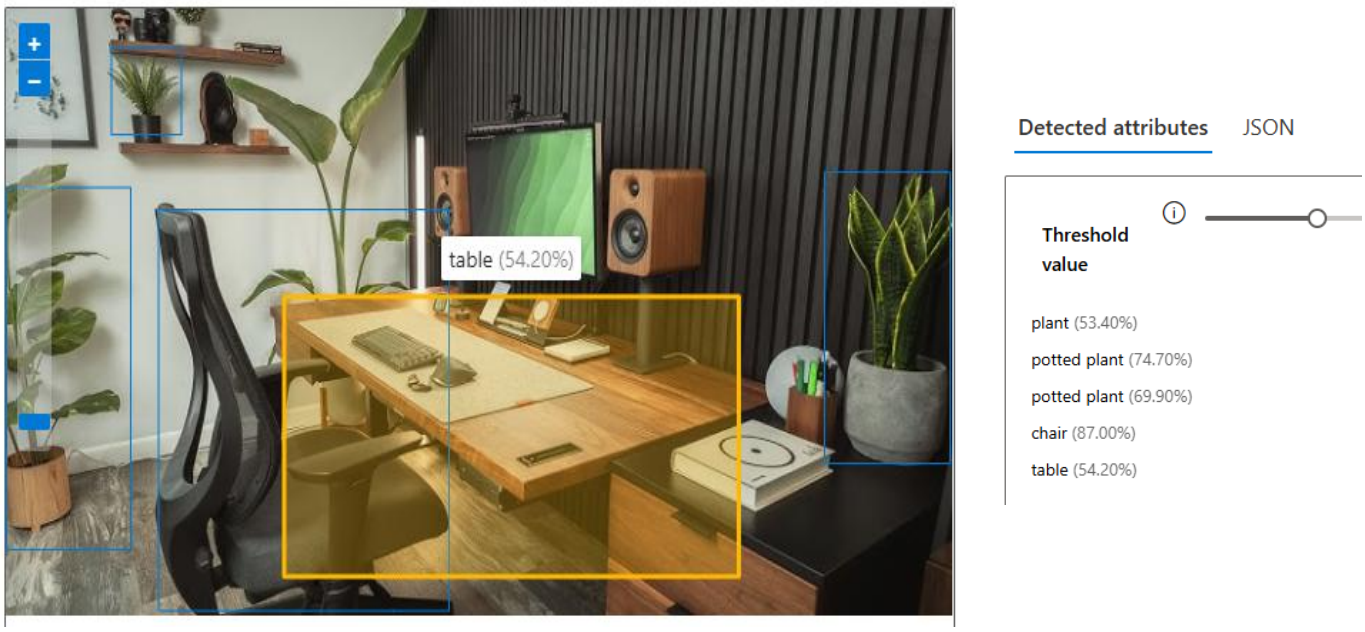
Azure AI Vision was used to analyze three distinct photos in order to assess the object detection system's performance. Every graphic illustrates a distinct detection situation that serves to highlight the model's advantages and disadvantages.

3.1 Image 1 – Workspace Environment

Figure 1: The original image of the workspace utilized to detect objects.

Source: A stock photo from websites that are accessible to the general audience.

Figure 2: Bounding boxes surrounding discovered objects are displayed in the object detection results.



In the first experiment, the object detection tool was used to analyze an image of a workplace. A plant, two potted plants, a chair, and a table were among the things in the scene that the system successfully identified. A bounding box was used to highlight each identified object, and each was given a confidence score that indicated the likelihood of a proper classification. The potted plants were identified with confidence levels of roughly 74.7% and 69.9%, respectively, while the chair received the highest confidence score of almost 87% among the detected objects. The plant and table were found with moderate confidence levels that were just above 50%.

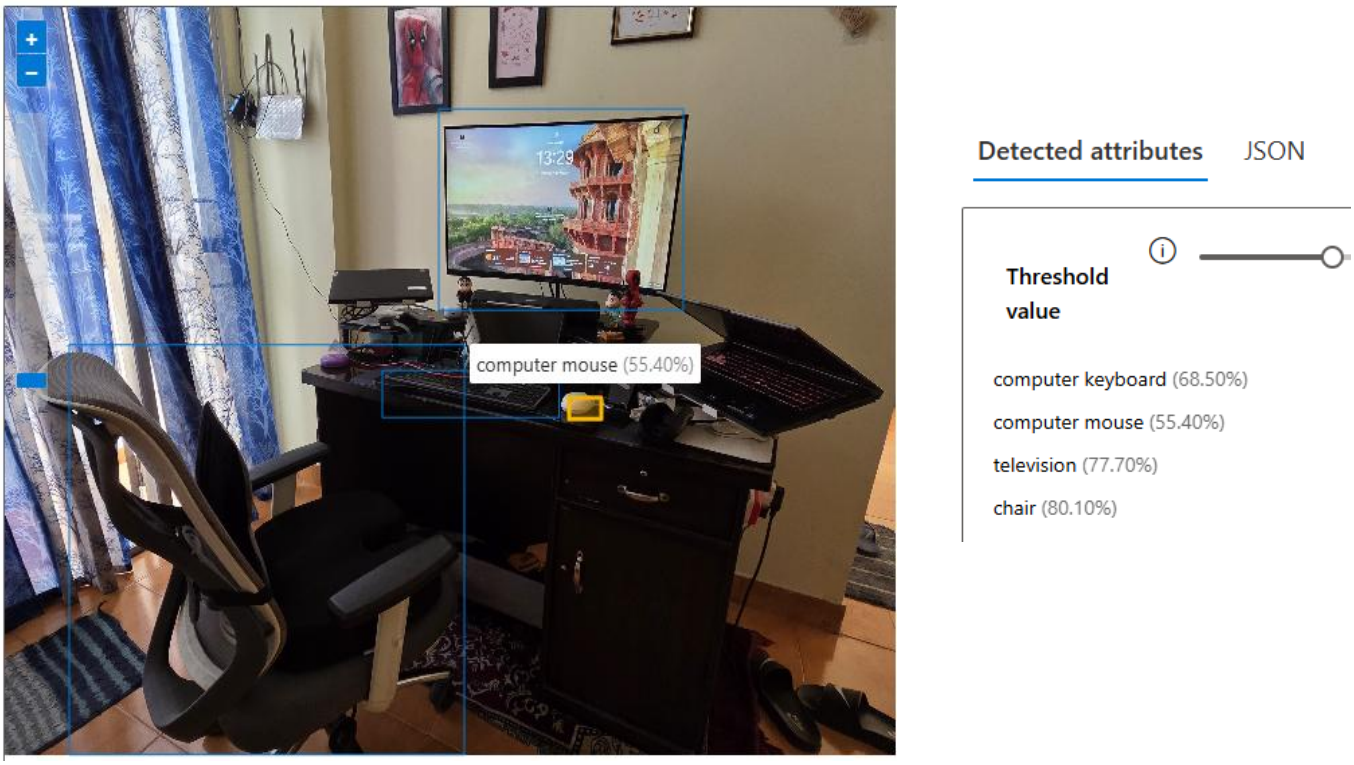
These findings show that the detection model does a good job of analyzing typical things that are easily observable in an organized setting. The well-separated objects in the workstation scenario take up a significant amount of the image, which probably helps explain the effective detection outcomes. The system's capacity to recognize recurring object categories within a single image was further demonstrated by its ability to classify several plants as distinct objects.

3.2 Image 2 – Personal Desk Setup

Figure 3: A test image of the initial desk configuration.

Source: The image of the author.

Figure 4: Object detection results for the personal desk setup.



In the second experiment, the Azure AI Vision object detection tool was used to analyze a picture of a personal desk arrangement. A computer keyboard, computer mouse, television, and chair were among the items in the scene that the system was able to identify. A bounding box was used to highlight each identified object, and each one was given a confidence score that represented the likelihood of a proper classification.

The chair had a high confidence score of about 80.1% of the objects that were spotted, suggesting that it had great detection reliability. With a confidence score of about 77.7%, the television—which matches the computer monitor in the picture—was identified. This classification shows that due to visual resemblance amongst display devices, the object detection model occasionally classifies monitors under the more general "television" label.

Additionally, the computer mouse and keyboard were identified with confidence levels of roughly 55.4% and 68.5%, respectively. The comparatively smaller size of these things in the picture or partial occlusion brought on by other desk items could be the source of these somewhat lower confidence values.

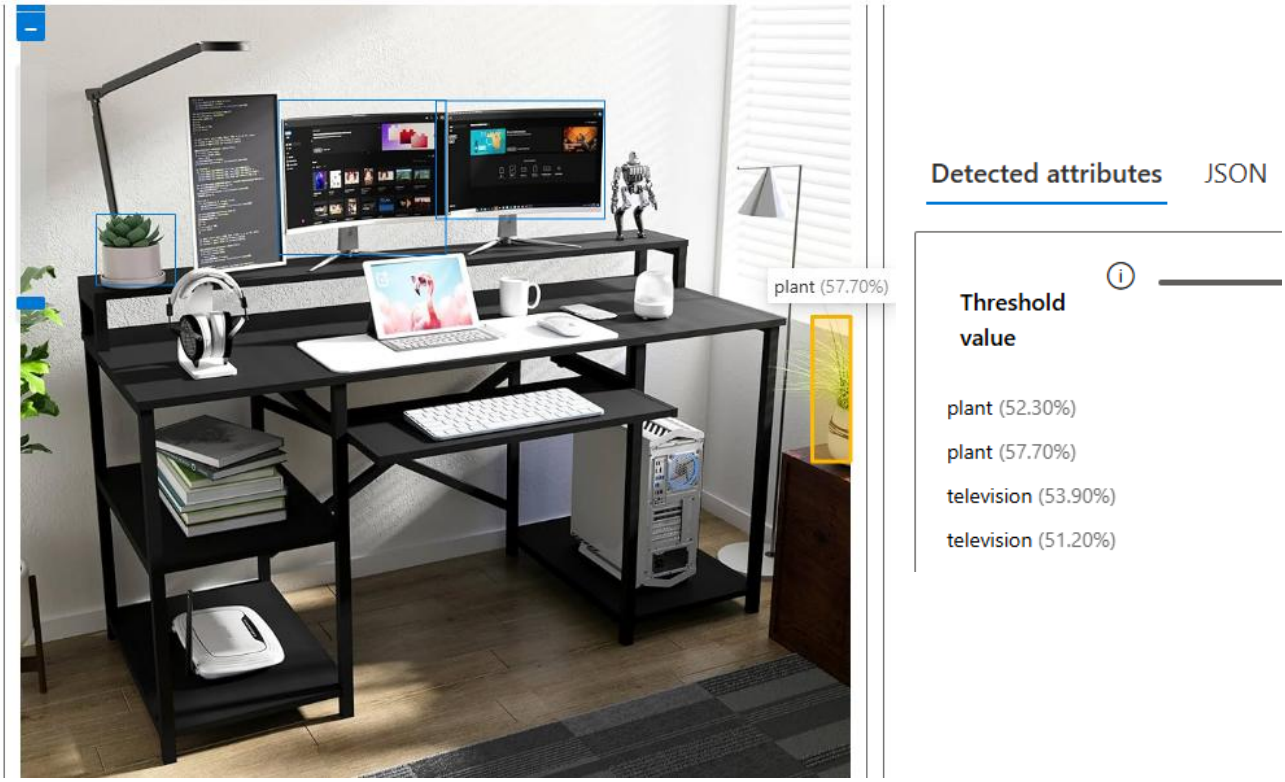
All things considered, the detection results demonstrate that the system can recognize several objects in a real-world workplace setting. The experiment does, however, also demonstrate how item size and visibility can affect detection confidence and how categorization may generalize seemingly related gadgets.

3.3 Image 3 – Duplicate Detection Scenario

Figure 5: Duplicate detection tests were conducted using the original desk scene.

Source: A stock photo from websites that are accessible to the general audience.

Figure 6: Duplicate object categories in detection results.



In the third experiment, Azure AI Vision's object detection feature was used to analyze a different working desk environment. The algorithm recognized a number of items in the picture, such as televisions and plants. According on the detection findings, the model identified two televisions with confidence values of roughly 53.9% and 51.2% and two plants with confidence scores of roughly 57.7% and 52.3%.

The two computer monitors on the desk match the televisions seen in the picture. The system categorized the monitors as televisions, just like in the prior trial. This suggests that the object identification model uses visual similarities to group various display devices into a more general television category.

The existence of duplicate object classifications in this experiment is an intriguing finding. Several instances of the same object categories were found in the scene by the model. Each of the two plants and two display screens in this instance has its own bounding box and confidence score. The model's capacity to identify recurring object classes within the same image is demonstrated by this behaviour .

In contrast to several earlier detections, the confidence levels for these detections are comparatively mild, just over 50%, indicating that the model had less conviction. The scene's complexity, object placement, and lighting can all have an impact on this.

Overall, this experiment highlights the model's propensity to generalize specific device kinds, like monitors, under more generic item categories, like televisions, while also demonstrating how the object detection system can identify many instances of comparable things inside a workplace setting.

4. Object Tracking Across Frames

The idea of object tracking in a real-world setting was illustrated using a brief film of a home office workspace. A desk, chair, monitor, and various surrounding items make up the computer workstation setup in the scenario. A person enters the frame and engages with the workstation while the video is being recorded. Three frames were taken at various points in the film to examine how object detection varies over time. After that, each frame was examined using Azure AI Vision via Vision Studio, which automatically recognizes items in the scene and highlights them with bounding boxes and confidence scores. It is feasible to track the appearance and movement of items in the environment over time by comparing the detection findings across these frames.

The retrieved frames show how Azure AI Vision reliably identifies objects even when there is movement in the scene. The system starts recognizing things in the surroundings as the person enters the workplace area in the first frame. The individual is seated at the workstation in the ensuing frames, but as they interact with the scene, their posture shifts. The system's bounding boxes adapt to the observed person's position across frames. This illustrates how mobility and environmental changes can be observed by comparing object detection results across several frames.

Analyzing subsequent frames offers a straightforward method to show how things emerge and move across a brief video sequence, even if Vision Studio's primary function is object detection rather than full continuous tracking.

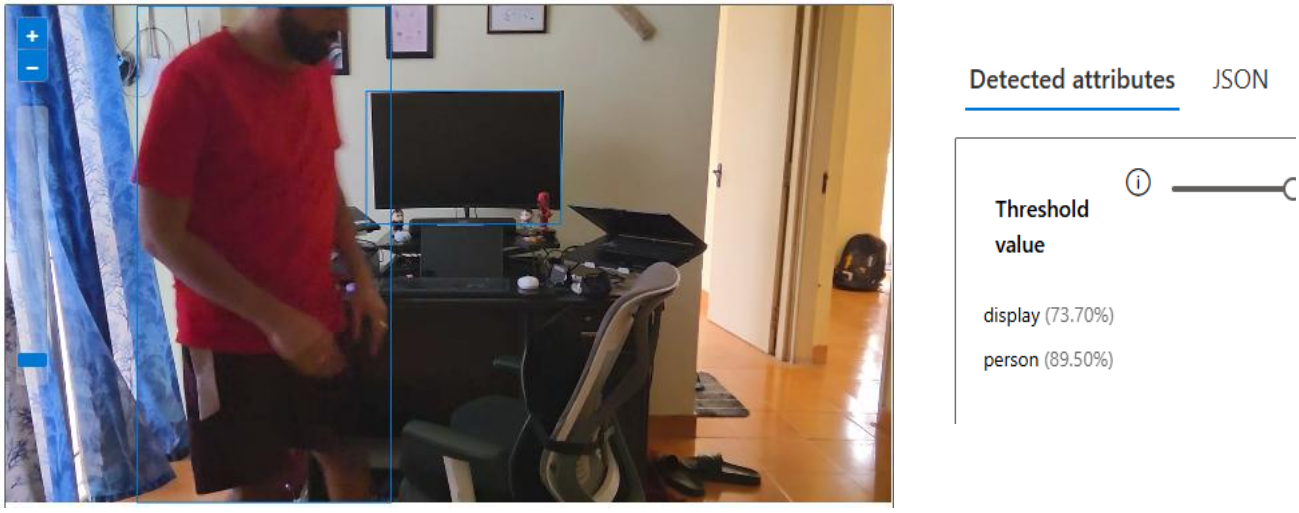
4.1 Frame 1 – Person entering the scene

Figure 7: Original Frame 1 from recorded video

Caption:

The individual approaching the workstation near the computer desk is seen in the opening frame.

Figure 8: Azure AI Vision detection result for Frame 1



Caption:

Azure AI Vision can identify objects in the scene, such as the chair, monitor, and person. Bounding boxes emphasize the objects in the workspace that have been identified.

The detection output for Frame 1 shows that Azure AI Vision successfully identifies the person entering the scene as well as the display (monitor) located on the workstation. The bounding boxes highlight the detected objects along with confidence scores, where the person is detected with a higher confidence level (approximately 89.5%) compared to the display object. This indicates that the model is more

confident in recognising human presence within the scene. The detection of the monitor as a display object also demonstrates the model's ability to recognise common workstation equipment within the environment.

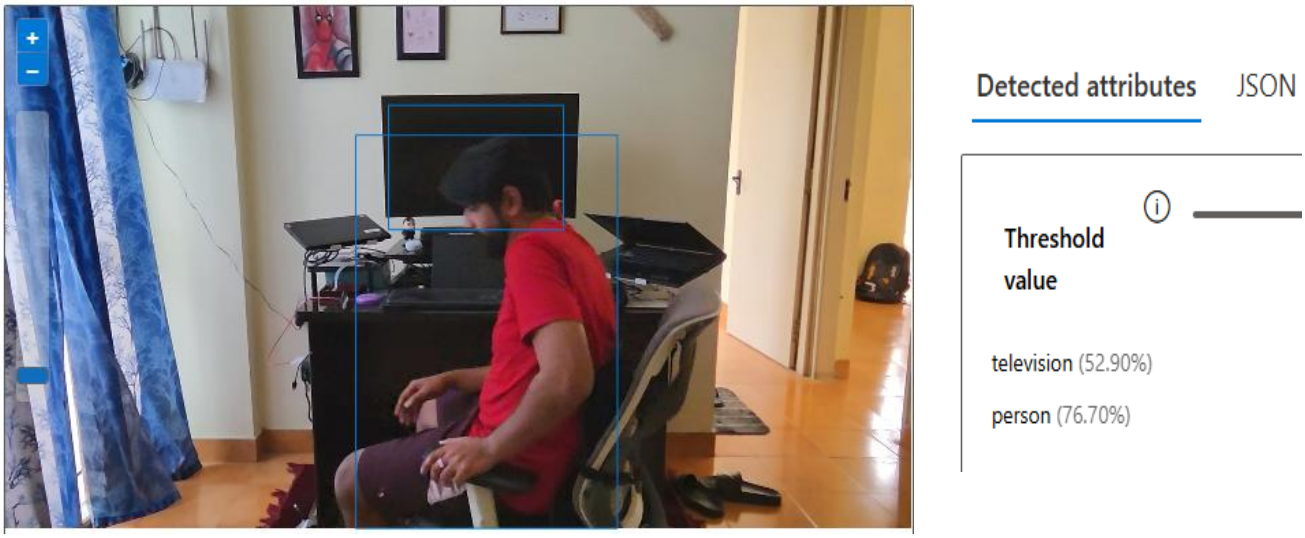
4.2 Frame 2 – Person seated at workstation

Figure 9: Original Frame 2 from recorded video

Caption:

The individual seated at the desk is shown interacting with the surroundings in the second frame.

Figure 10: Azure AI Vision detection result for Frame 2



Caption:

While the individual is seated, Azure AI Vision keeps detecting them as well as nearby items like the chair and monitor.

The individual seated at the workstation after entering the scene is seen in the detection output for Frame 2. With a confidence score of roughly 76.7%, Azure AI Vision is still able to identify the human as the main object in the frame. Additionally, with a confidence score of about 52.9%, the algorithm recognizes the monitor as a television/display item. The locations of these identified objects in the surroundings are shown by the bounding boxes. The identified individual has shifted from standing to sitting in the scene in comparison to the preceding frame. This illustrates how changes in object position over time can be observed using object detection findings across numerous frames.

4.3 Frame 3 – Person posture change

Figure 11: Original Frame 3 from recorded video

Caption:

The individual is shown shifting their posture while seated at the desk in the third frame.

Figure 12: Azure AI Vision detection result for Frame 3

Caption:

While the surrounding workspace objects stay largely motionless, the detection output emphasizes the human as the main object in the scene.

The person seated at the workstation is shown in the detection result for Frame 3 waving their hand in the direction of the camera. With a confidence score of almost 87.9%, Azure AI Vision recognizes the human as the main object in the frame, which is higher than the previous frame. The bounding box envelops the individual and modifies based on the updated hand movement and posture. This frame shows how the identified object's position and pose change over time, yet it stays constant between frames. The system's capacity to consistently recognize the same item in spite of postural alterations demonstrates how object detection outcomes can be progressively observed to comprehend movement inside a scene.

5 Performance Evaluation

The capacity of Azure AI Vision to accurately identify items throughout the testing photos and video frames was used to assess its performance. The system was able to identify a number of common objects in the workstation surroundings, including tables, seats, plants, monitors, and keyboards, in the image-based studies. Bounding boxes and confidence ratings, which show how certain the model is in identifying each object, were displayed alongside the detection results.

The findings demonstrated that the model does a good job of identifying objects that are easily observable and well recognized. For instance, across several photos and frames, objects like chairs, monitors, and individuals were consistently identified with comparatively high confidence scores. This suggests that Azure AI Vision is capable of identifying common objects in indoor settings.

Nevertheless, certain restrictions were also noted. Certain things that were clearly apparent in certain photos were not detected. For instance, the model occasionally failed to detect tiny objects or partially obscured objects. Furthermore, some items were sometimes categorized under more general terms, like identifying a monitor as a television or display device.

Even when the person's posture and position varied during the video frame analysis, the model was still able to reliably identify them. This illustrates how the model can continue to recognize objects consistently across successive frames. Overall, the findings show that Azure AI Vision has dependable object detection capabilities, especially for everyday objects and settings that are easy to see.

6 Strengths and Limitations of Azure AI Vision

For object detection jobs, Azure AI Vision offers a number of benefits. Its simplicity of use through Vision Studio, which enables users to input photographs and analyze detection findings without having in-depth programming knowledge, is one of its main advantages. It is simple to understand the recognized items in a scene thanks to the platform's clear visual outputs, which include bounding boxes and confidence scores.

The system's capacity to identify a broad range of everyday things is another asset. Azure AI Vision was able to identify objects including people, chairs, plants, monitors, keyboards, and tables during the tests. In the video experiment, the model also showed consistent detection across several frames, suggesting trustworthy identification of moving items in a scene.

Despite these advantages, certain drawbacks were noted. Even if they were present in the picture, certain things were not picked up, especially if they were little, partially concealed, or situated in congested areas. Furthermore, rather than being categorized under particular object kinds, certain objects were placed under broader categories. For instance, computer displays were occasionally mistaken for televisions or other display devices.

These findings show that although Azure AI Vision works well for common object detection tasks, image complexity, object size, and contextual factors can affect how well it performs.

7 Conclusion

This experiment used both static photos and extracted video frames to show how to use Azure AI Vision for object detection and basic object tracking. The system's capacity to identify frequent objects in indoor scenarios was assessed by analyzing many photos of workplace settings. According to the findings, Azure AI Vision was able to identify a number of things with respectable confidence levels, including tables, chairs, plants, monitors, and keyboards.

Object tracking was demonstrated by analyzing several frames taken from a brief video capture of a home workplace, in addition to image-based detection. A person entering the location, sitting at the desk, and interacting with the surroundings were all captured on camera. It was feasible to see how the bounding box surrounding the identified individual altered while movement took place within the scene by comparing detection findings between frames.

All things considered, the experiment shows that Azure AI Vision is a helpful tool for identifying and evaluating items in pictures and video frames. The system can be used in a variety of real-world applications, including automated visual analysis, security systems, and workspace monitoring, and it offers an easily accessible platform for developing computer vision capabilities.

8 Future Improvements

Even though Azure AI Vision showed good object detection capabilities, the experiment might be improved in a few ways. Analyzing a wider range of photos with more varied settings and item kinds is one potential enhancement. A more thorough assessment of the system's performance may be obtained by testing the model in various lighting scenarios, camera angles, and scene complexity levels.

Analyzing lengthier video sequences with more continuous movement would be another enhancement. Future research could examine several frames from a larger video clip rather than just three extracted ones in order to better observe object movement and assess tracking behaviour over time.

Additionally, integrating the Azure AI Vision API within a programming environment could allow for automated detection across multiple images and frames. This would enable more advanced analysis, such as continuous object tracking, real-time detection, and automated processing of larger datasets.

These improvements would help provide a deeper understanding of the capabilities and limitations of Azure AI Vision in real-world computer vision applications.